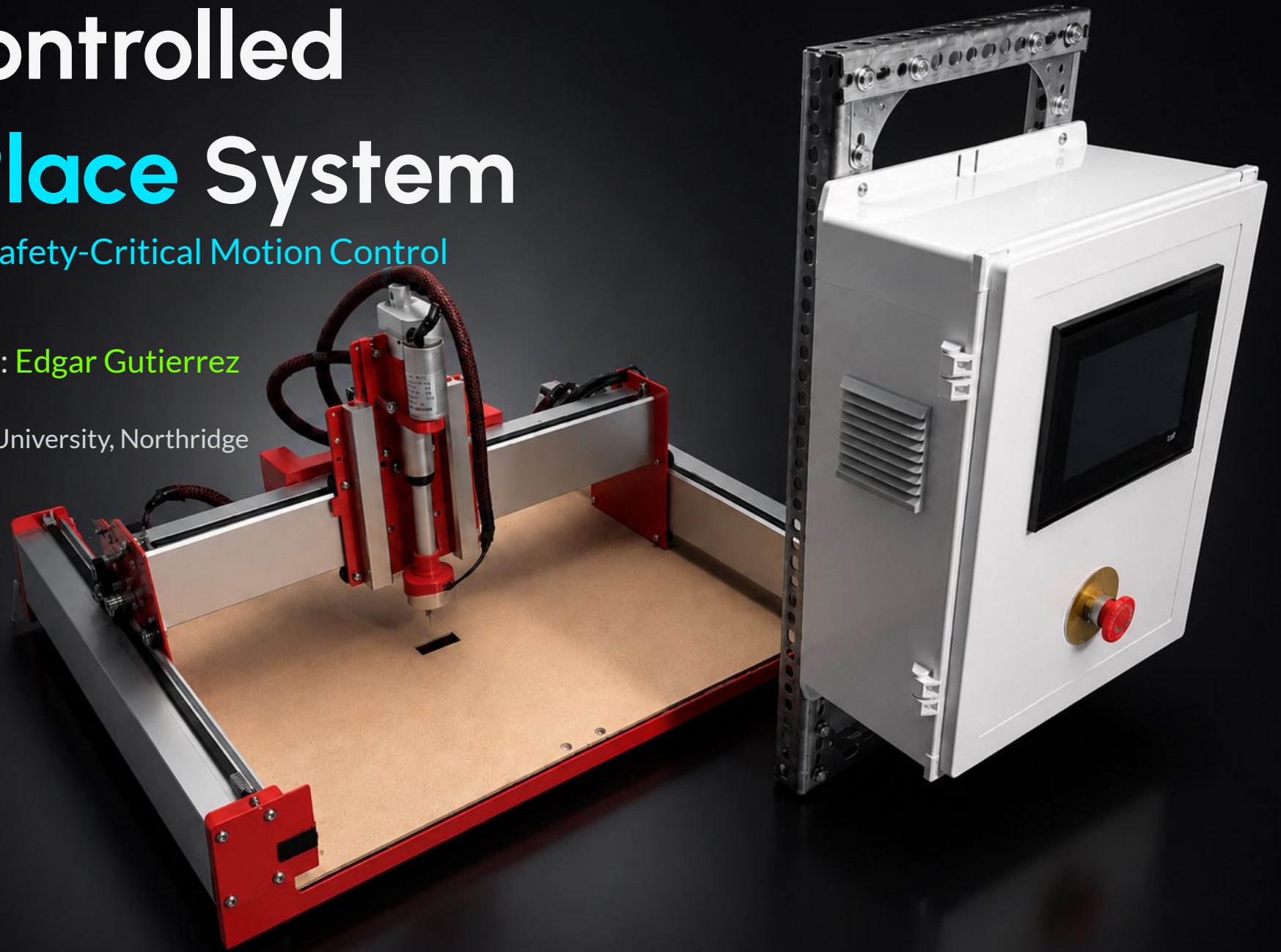


PLC-Controlled Pick-and-Place System

Hardware Integration & Safety-Critical Motion Control

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INDIVIDUAL PROJECT OBJECTIVES



Hardware Integration

Transitioning hobbyist hardware to professional industrial standards using the LS Electric XBC PLC.



Industrial Safety

Implementing hardware-level interlocking and UL 508A compliant enclosure layouts.



Power Strategy

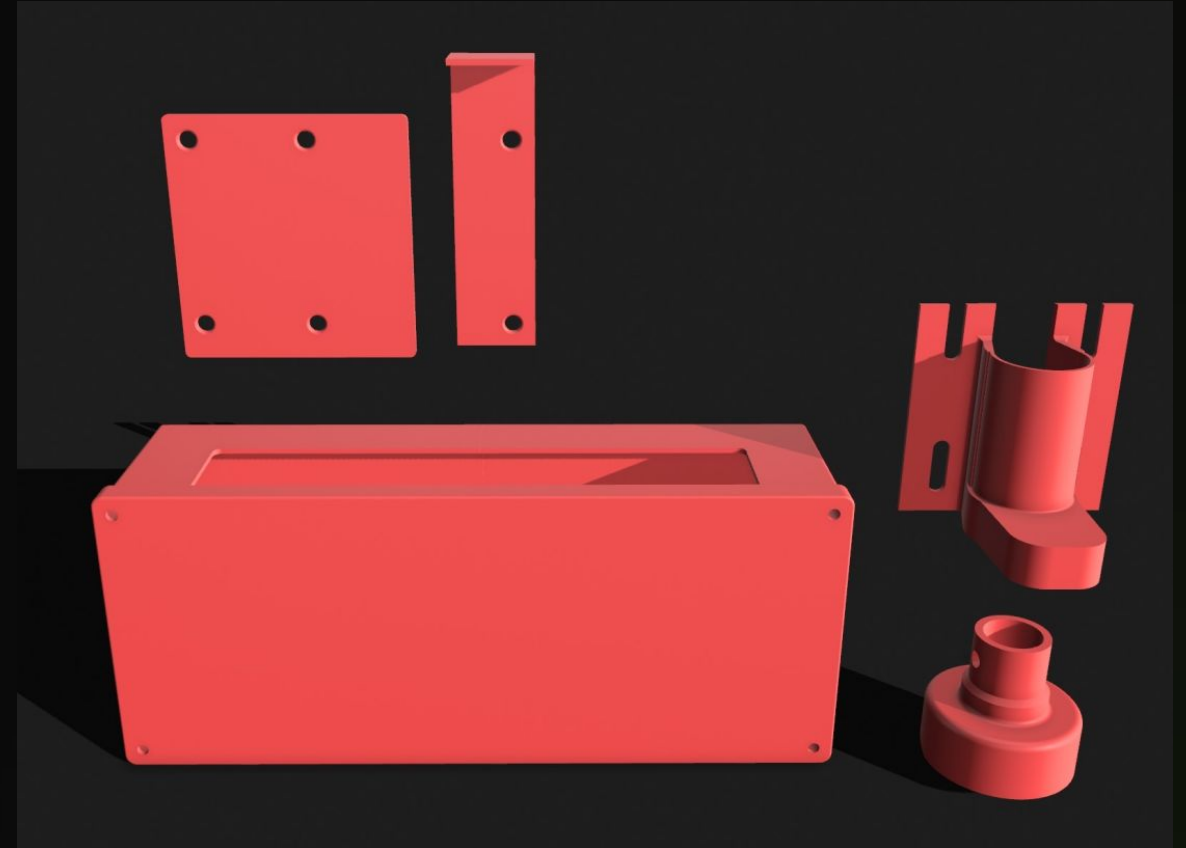
Establishing a dual-rail 24V DC distribution system to isolate motion noise from logic.

| MECHANICAL & CUSTOM INTEGRATION

Custom PETG Modifications

Engineered and 3D printed high-density PETG mounts to adapt the Shapeoko frame for industrial use.

-  **End-Effector Mounts:** Coupling linear actuator and electromagnet.
-  **Limit Switch Brackets:** Hard-wired overtravel protection.
-  **Gantry Junction Box:** Localized termination to minimize motion-crossing cables.



ELECTRICAL HARDWARE STANDARDS

- ✓ **UL 508A Compliance:** Designed the control enclosure to follow industrial safety standards for power and signal separation.
- 🔧 **NEC Article 409:** Strategic component spacing on metal back panels for grounding and rigid mounting.
- 🎨 **Industrial Color Coding:** Blue (24V DC), Red (Output), White (Common), Black/White (120V AC).
- 🏷️ **Traceable Wiring:** All I/O points consolidated through labeled landing terminals for rapid troubleshooting.

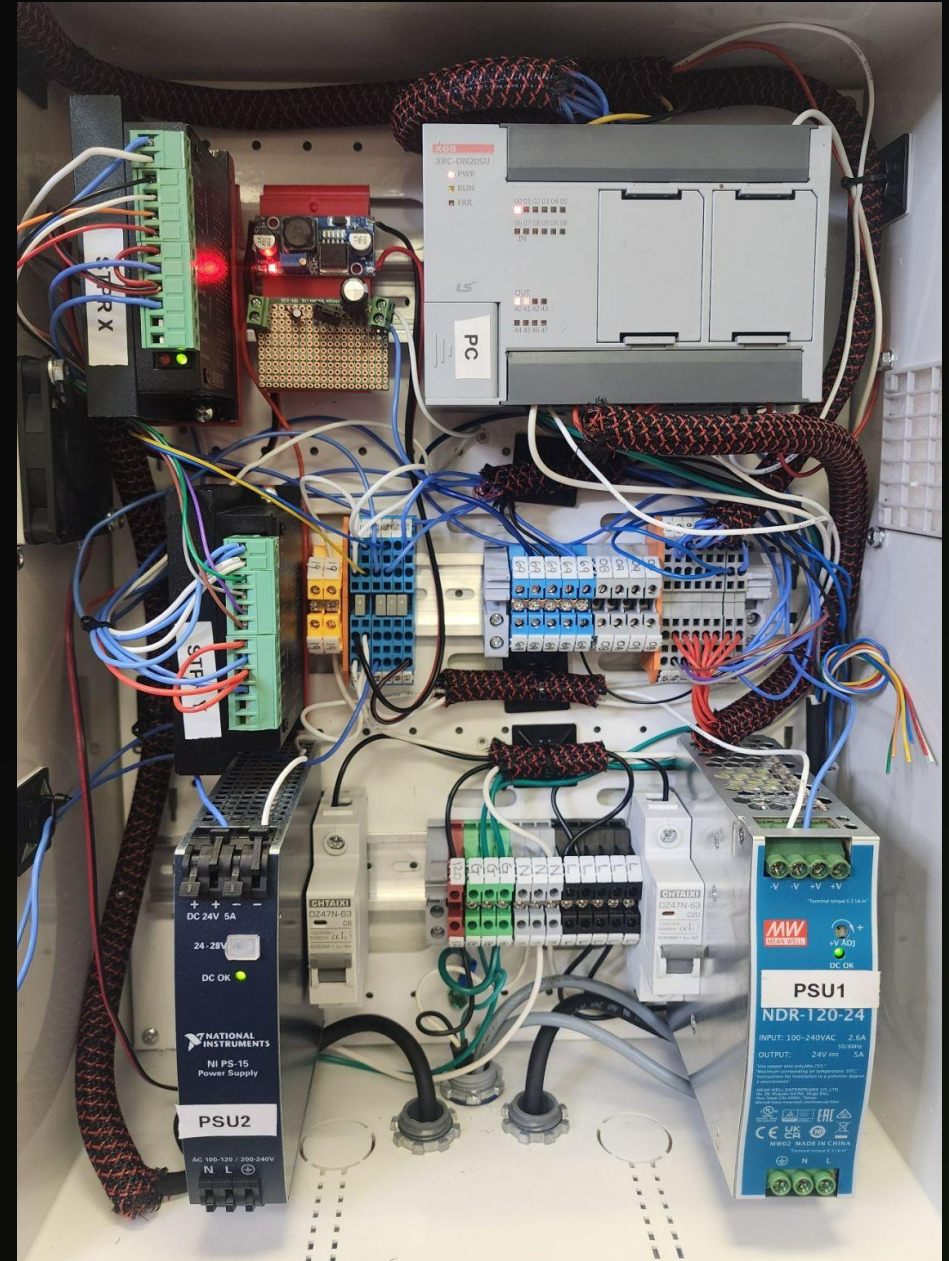
CONTROL PANEL

Infrastructure Design

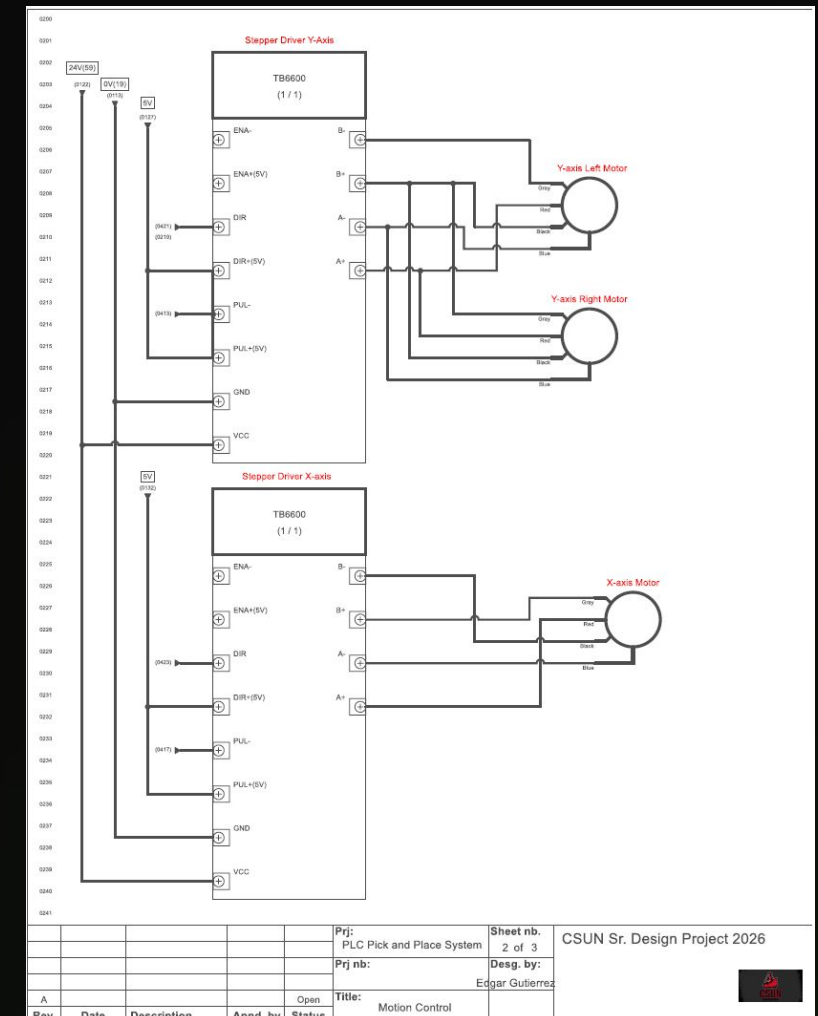
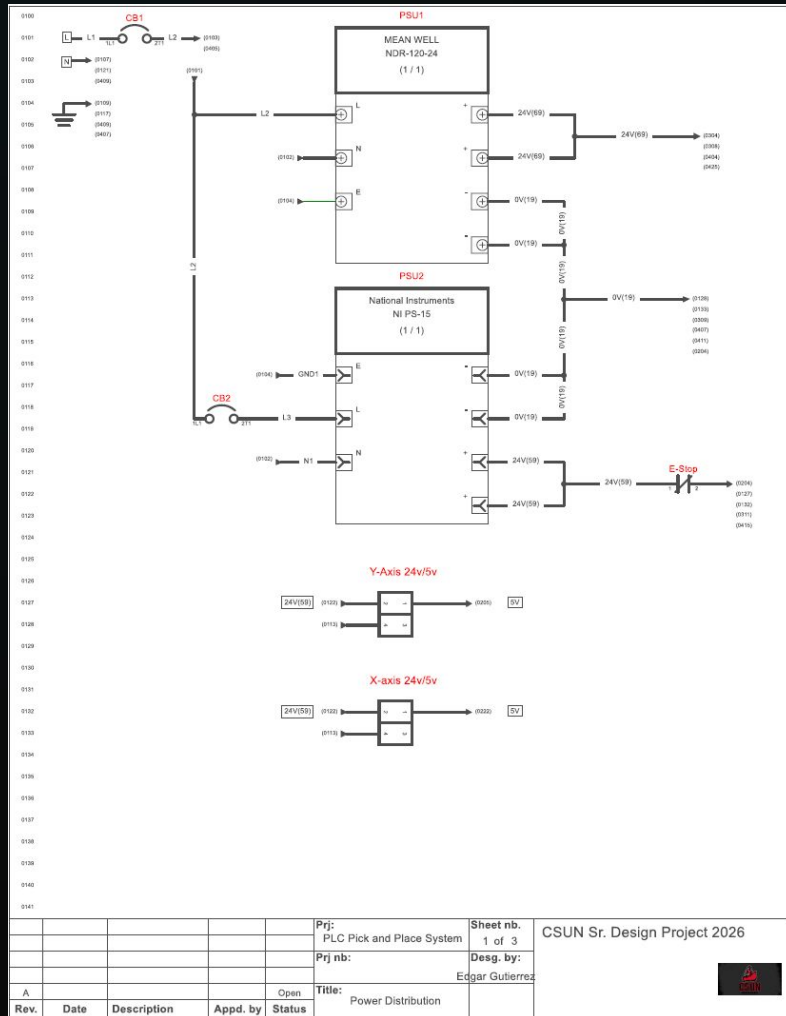
The internal architecture prioritizes modularity. Dual DIN rails house power distribution, circuit breakers, and motion drivers.

The **LS Electric XBC-DN20SU** is positioned for optimal I/O routing to the terminal blocks.

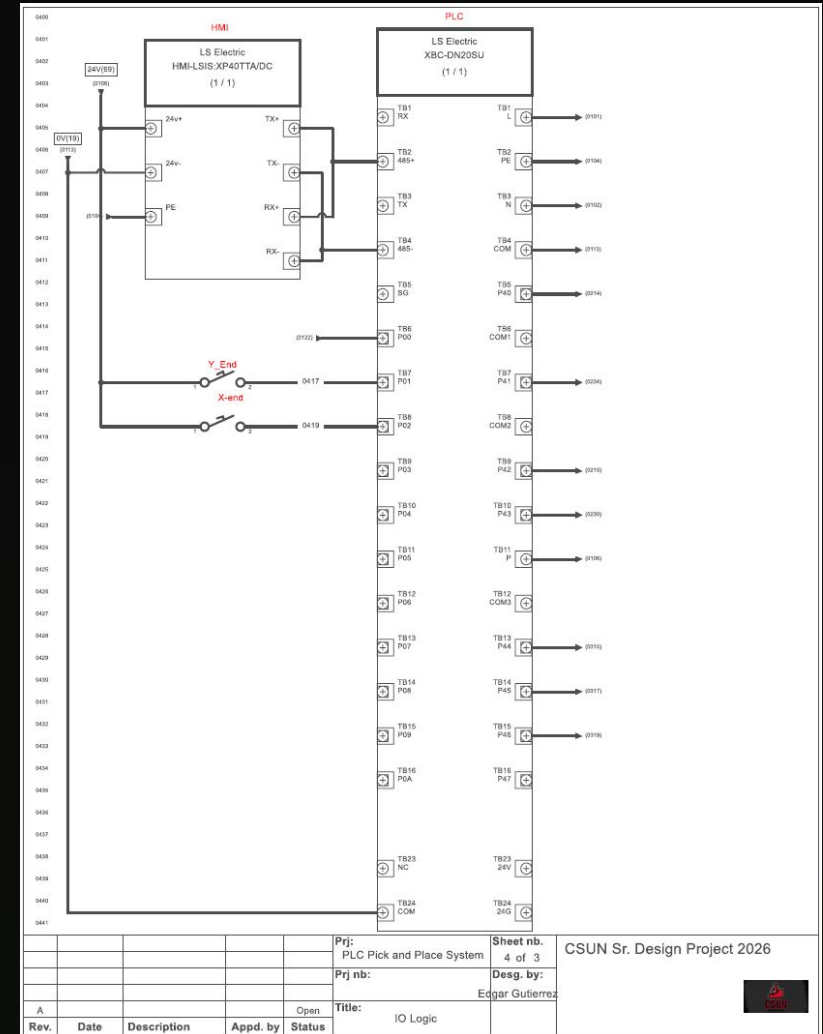
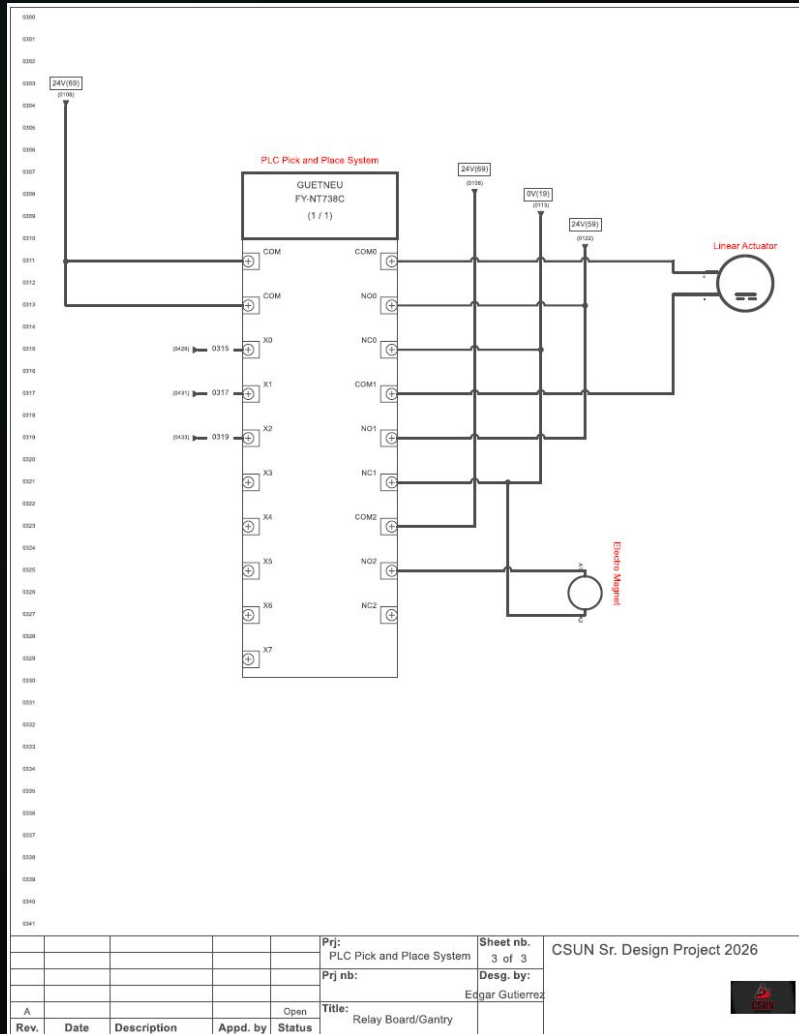
Shielded cable routing mitigates EMI between high-current stepper power and sensitive signal wires.



SCHEMATIC/WIRING DIAGRAM



SCHEMATIC/WIRING DIAGRAM



SYSTEM POWER BUDGET ANALYSIS

Component	Voltage	Peak Current	Power (Watts)
Stepper Motors (X, Y) - PSU2	24V DC	3.00 A	72 W
Linear Actuator (Z) - PSU2	24V DC	3.00 A (Stall)	72 W
Electromagnet - PSU1	24V DC	0.25 A	6 W
X and Y Limit Switches - PSU1	24V DC	0.50 A	12 W
TOTAL SYSTEM PEAK	24V DC	6.75 A	162 W

Dual 120W Power Supplies utilized for load segregation and motion noise isolation.

| MOTION PRECISION METRICS

0.012

mm per step

High-Resolution Control

Achieved through TB6600 driver micro-stepping and PLC high-speed pulse output (PTO) configuration.

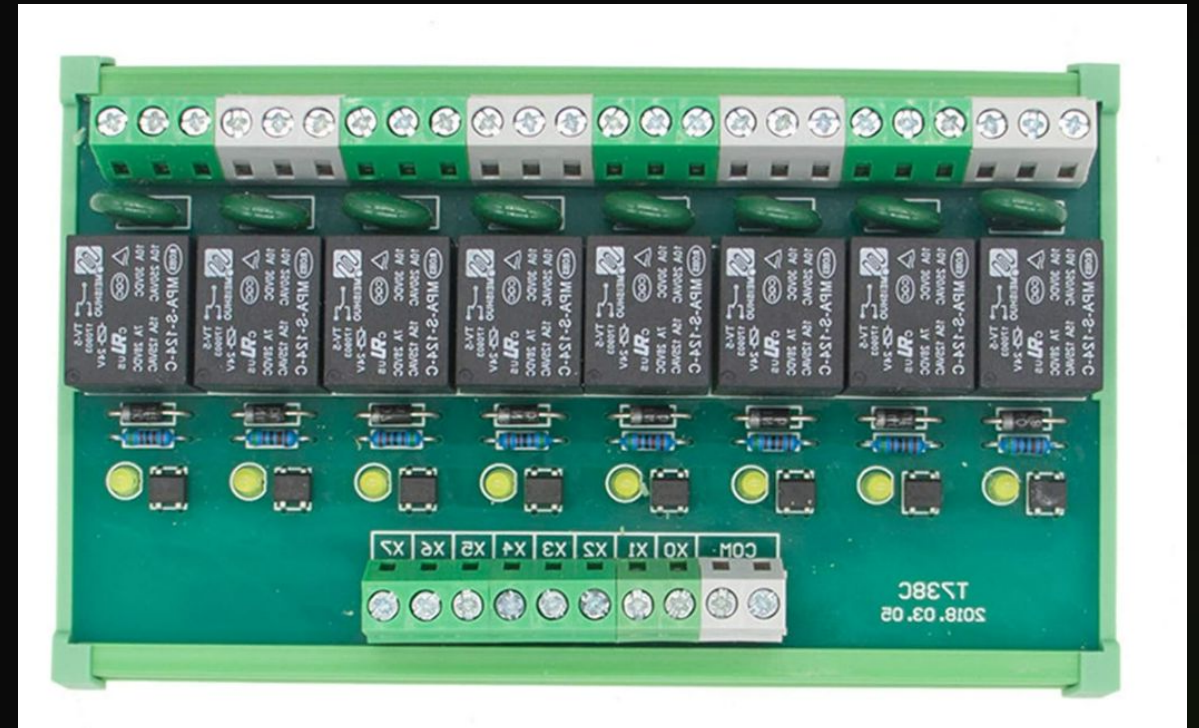
Validated via repeatable pick-and-place cycles on a custom grid array.

CHALLENGE: SINKING OUTPUT LOGIC

Signal Logic Refinement

The LS Electric PLC uses **NPN (Sinking)** outputs. This created a mismatch with standard relay boards designed for sourcing logic.

Solution: Re-engineered the relay board power distribution to utilize a common +24V DC rail, allowing the PLC to complete the circuit to 0V (Common).



PROJECT EXECUTION MILESTONES

Planning

Hardware sourcing & benchmarks.

Logic Dev

PTO mapping & baseline routines.

Integration

Wiring, 3D printing & enclosure build.

Validation

Functional cycles & safety testing.

Thank You!

PLC-Controlled Pick-and-Place
System

Questions?

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